

Design Optimization of Carbon/Epoxy Laminates under Stiffness and Strength Constraints

Eesha Tolety ^{*1} and Aditya K. Singh ^{†2}

^{1,2}Ann and H.J. Smead Aerospace Engineering Sciences, University of Colorado, Boulder

May 10, 2025

Abstract

This project focuses on the design and optimization of a composite laminate aimed at achieving the minimum possible weight while satisfying specific stiffness and strength criteria. The laminate design process involves determining the number of plies, their orientation angles, and maintaining a uniform fiber volume fraction within a specified range. Utilizing the properties of carbon/epoxy lamina, the design approach unfolds in three stages: (1) optimizing the ABD matrix to meet stiffness constraints, (2) ensuring structural integrity under various loading scenarios by applying the Tsai-Wu failure criterion, and (3) simultaneously addressing both stiffness and strength conditions. The final design strikes a balance among competing requirements by systematically exploring the design space, assessing mechanical responses, and refining the laminate configuration. The findings underscore the trade-offs between stiffness and strength, illustrating the methodology used to justify the final chosen solution.

^{*}eeto3049@colorado.edu

[†]Aditya.Singh-2@colorado.edu

1 Introduction

Composite materials, especially carbon fiber-reinforced polymers, are renowned for their high stiffness-to-weight and strength-to-weight ratios, making them highly suitable for aerospace structural applications. This project, part of Composite Structures and Materials, aims to design the lightest possible laminate that meets a specified set of mechanical performance criteria. The objective of this work is to develop efficient structural solutions that meet the concurrent requirements of stiffness and strength, which are essential characteristics of advanced aerospace components.

The design space encompasses laminates with varying numbers of plies, diverse ply orientations, and a consistent fiber volume fraction ranging from 45 to 65 percent. The constituent materials—carbon fibers and an epoxy matrix—possess well-defined mechanical and failure properties. The project is divided into three distinct phases. The first phase concentrates exclusively on stiffness, requiring the in-plane (A) and bending (D) matrices of the laminate to exceed defined threshold values across all orientations. In the second phase, the laminate's capacity to endure complex loading conditions is evaluated using the Tsai-Wu failure criterion, excluding explicit safety factors. The final phase combines both stiffness and strength criteria, presenting the designer with the challenge of identifying a configuration that satisfies all mechanical requirements while minimizing weight.

This report details the design process, covering material modeling, the application of laminate theory, failure analysis, and strategies for weight minimization. The methodology highlights the relationship between design decisions and performance metrics, ultimately illustrating a systematic approach to optimizing composite laminates under multiple constraints.

2 Methodology

The laminate design process was organized into two primary phases: (1) ensuring stiffness criteria by analyzing the ABD matrix, and (2) verifying strength using the Tsai-Wu failure criterion across different load combinations. We used MATLAB Code to do all the required calculations and help optimize the layup configuration.

2.1 Design for Stiffness

The stiffness aims to find the lightest laminate that meets the following requirements mentioned in the Problem statement:

- $A_{xx} \geq 100,000 \text{ N/mm}$ in the global x-y reference frame
- $A_{\bar{x}\bar{x}} \geq 30,000 \text{ N/mm}$ for any rotated x-y frame
- $D_{xx} \geq 25,000 \text{ N} \cdot \text{mm}$

- $D_{yy} \geq 15,000 \text{ N} \cdot \text{mm}$

ABD matrix [1] needs to be calculated to design the composite. It is a 6×6 matrix with components for the k^{th} defined as :

$$A_{ij} = \sum_{k=1}^n Q_{ij}^k (z_{k+1} - z_k) \quad (1)$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^n Q_{ij}^k (z_{k+1}^2 - z_k^2) \quad (2)$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^n Q_{ij}^k (z_{k+1}^3 - z_k^3) \quad (3)$$

Arranging equations 1 - 3 in matrix form Like

$$\begin{bmatrix} A & B \\ B & D \end{bmatrix} \quad (4)$$

gives us the ABD matrix where each subcomponent is a 3×3 matrix in itself. As indicated in the equations (1 - 3), we need the stiffness matrix Q , and $(z_{k+1} - z_k)$, can be obtained by selecting the volume fraction and thickness of each lamina.

To find the volume fraction V_f , and thickness of the lamina, we are given a reference case which is that when $V_f = 50\%$ then the thickness is 0.1 mm. Using this reference case, we can find the thickness for different V_f as :

$$t = 0.1 \times \frac{0.50}{V_f} \quad (5)$$

Now, we need to calculate the stiffness (Q) matrix, which is given as :

$$\begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \quad (6)$$

where the respective entries are:

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}}$$

$$Q_{12} = Q_{21} = \frac{\nu_{21}E_2}{1 - \nu_{12}\nu_{21}}$$

$$Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}}$$

$$Q_{66} = G_{12}$$

Elements of the matrix can be evaluated using the Simple Rule of Mixtures and Halpin-Tsai relations [2] :

$$G_m = \frac{E_m}{2(1 + \nu)} \quad (7)$$

$$E_1 = E_{1f}V_f + E_m(1 - V_f) \quad (8)$$

$$E_2 = E_m \frac{1 + \xi_1 \eta_1 V_f}{1 - \eta_1 V_f} \quad (9)$$

$$\eta_1 = \frac{E_{2f} - E_m}{E_{2f} + \xi_1 E_m} \quad (10)$$

$$\nu_{12} = \nu_{12f}V_f + \nu_m(1 - V_f) \quad (11)$$

$$\nu_{21} = \nu_{12}E_2 \quad (12)$$

$$G_{12} = G_m \frac{1 + \eta_2 \xi_2 V_f}{1 - \eta_2 V_f} \quad (13)$$

$$\eta_2 = \frac{G_{12f} - G_m}{G_{12f} + \xi_2 G_m} \quad (14)$$

As per the given input data the Fiber Properties vector is arranged as :

$$[E_{1f} \ E_{2f} \ G_{12f} \ \nu_{12f}] = [270 \ (GPa) \ 14 \ (GPa) \ 10 \ (GPa) \ 0.22]$$

Similarly, we have Matrix Properties vector:

$$[E_m \ \nu_m] = [3.5 \ (GPa) \ 0.37]$$

and Composite Properties vector :

$$[V_f \ \xi_1 \ \xi_2] = [range(0.4 - 0.65) \ 1 \ 1]$$

We chose the system of units as *MPa and mm*. Using the above-mentioned values, we can calculate the *Q* matrix (6) in the 1-2 frame, which is the frame aligned with the principal direction of fibres present in individual laminas. But, to obtain the *Q* matrix in the x-y reference frame, we use the transformation of matrices concept from linear algebra and is defined as :

$$\begin{bmatrix} Q_{xx} & Q_{xy} & Q_{xs} \\ Q_{yx} & Q_{yy} & Q_{ys} \\ Q_{sx} & Q_{sy} & Q_{ss} \end{bmatrix} = T^{-1} \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} T \quad (15)$$

where *T* is the transformation matrix and is defined as :

$$\begin{bmatrix} \sin^2\theta & \sin^2\theta & 2 * \cos\theta * \sin\theta \\ \sin^2\theta & \sin^2\theta & -2 * \cos\theta * \sin\theta \\ -\cos\theta * \sin\theta & \cos\theta * \sin\theta & \cos^2\theta - \sin^2\theta \end{bmatrix} \quad (16)$$

and θ is the angle between the principal direction of lamina fibers and the x-y frame in an anticlockwise sense. Utilizing all these equations, we built functions for calculating the stiffness matrix Q and the ABD matrix. A *MATLAB* code was built using these functions and then used to get the values for the comparison parameters A_{xx} , $A_{\bar{x}\bar{x}}$, D_{xx} , and D_{yy} . We used this code to obtain the lamina stacking sequence or layup, which satisfied the stiffness design constraint values.

2.2 Strength design

In our pursuit to identify the lightest feasible laminate that can endure all outlined load conditions we utilized the *Tsai-Wu criterion*, an interactive failure theory, for failure prediction. The laminate is required to survive the following applied loads :

- $N_{xx} = 600 \text{ N/mm}$
- $N_{\theta\theta} = 200 \text{ N/mm}$
- $M_{xx} = \pm 100 \text{ N}$

where $N_{\theta\theta}$ varies along with θ angle from the x-axis. The strength of each lamina under loading was assessed using the *Tsai-Wu* failure theory, a quadratic interactive criterion that integrates the effects of normal and shear stresses to predict the onset of failure. For a two-dimensional state of stress aligned with the principal material directions, the Tsai-Wu failure index [1] is expressed as follows:

$$f_1\sigma_1 + f_2\sigma_2 + f_{11}\sigma_1^2 + f_{22}\sigma_2^2 + f_{66}\tau_6^2 + 2f_{12}\sigma_1\sigma_2 = 1 \quad (17)$$

where:

$$f_1 = \frac{1}{F_{1t}} - \frac{1}{F_{1c}} \quad (18)$$

$$f_2 = \frac{1}{F_{2t}} - \frac{1}{F_{2c}} \quad (19)$$

$$f_{11} = \frac{1}{F_{1t}F_{1c}} \quad (20)$$

$$f_{22} = \frac{1}{F_{2t}F_{2c}} \quad (21)$$

$$f_{66} = \frac{1}{F_6^2} \quad (22)$$

$$f_{12} = -\frac{1}{2}\sqrt{f_{11}f_{22}} \quad (23)$$

The failure terms in the equations 18 -22 can be found from the already available failure properties of the material as:

$$\mathcal{E}_f^u = 1\% \quad , F_{mt} = 120 \text{ MPa} \quad , F_{mc} = 150 \text{ MPa} \quad , F_{ms} = 80 \text{ MPa}$$

The relations between the Failure properties to the coefficients in the *Tsai-Wu* are given as :

$$F_{ft} = E_{1f} \times \varepsilon_f^u \quad (24)$$

$$F_{1t} = (F_{ft} \times V_f) + (F_{mt} \times (1 - V_f)) \quad (25)$$

$$K_2 = \frac{(1 - V_f) \times \left(1 - \frac{E_m}{E_{2f}}\right)}{1 - \left(\sqrt{\frac{V_f}{4\pi}}\right) \times \left(1 - \frac{E_m}{E_{2f}}\right)} \quad (26)$$

$$F_{2t} = \frac{F_{mt}}{K_2} \quad (27)$$

$$F_{1c} = G_{12} \quad (28)$$

$$K_\tau = \frac{(1 - V_f) \times \left(1 - \frac{G_m}{G_{12f}}\right)}{1 - \left(\sqrt{\frac{V_f}{4\pi}}\right) \times \left(1 - \frac{G_m}{G_{12f}}\right)} \quad (29)$$

$$F_6 = \frac{F_{ms}}{K_\tau} \quad (30)$$

$$F_{2c} = 1.2 \times F_{2t} \quad (31)$$

All the relations above are taken from [1]. Now, to find the Stress in 17 equation, we use the relation of the *ABD* matrix:

$$\begin{bmatrix} \varepsilon_0 \\ \kappa \end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix}^{-1} \begin{bmatrix} N \\ M \end{bmatrix} \quad (32)$$

Where ε_0 is the mid-plane strain κ is the curvature vector.

Then, $\sigma(z)$ is given as a $\overline{Q}^\theta \cdot \varepsilon(z)$. As the σ here is in the global stress, it needs to be transformed to the local coordinates. Here, the transformation is represented:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = T(\theta) \cdot \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}$$

Using the established formulas and relationships, code was developed to test stacking sequences that will withstand the specified loads.

Following our recent evaluations of the composite's strength and strain, our next step is to identify a lightweight configuration. To accurately assess the weight per unit square using the density of the composite.

$$W(\text{ per unit area}) = \rho_{\text{composite}}(\text{kg/m}^3) \times g(\text{m/s}^2) \times \text{thickenss}(\text{mm})$$

where:

$$\rho_{composite} = (\rho_f \times V_f) + (\rho_m \times (1 - V_f))$$

from the ratio of mixtures formula , and g is acceleration due to gravity ($=9.81$).

After careful consideration of all relevant information, we have selected a lamina stacking sequence that addresses strength design, strain design, and incorporates both design approaches.

3 Results

Three distinct laminate configurations were investigated during the design process, each tailored to a specific objective: optimizing stiffness, optimizing strength, and creating a combined stiffness-strength solution.

The initial configuration $[0\ 0\ 0\ 0\ 35\ 90\ 90\ 90\ 90]_s$ is developed specifically to meet the stiffness requirements. In this setup, the volume fraction is set at **0.4**, and the thickness of each lamina is **0.125 mm**. The total weight of the composite is **27.8 kg /m²**. By employing the ABD matrix as to compare our results with the evaluation metrics, we got the values as :

- $A_{xx} = 1.0207 \times 10^5$ N/mm in the global x-y reference frame
- $\min(A_{\bar{x}\bar{x}}) = 8.2686 \times 10^4$ N/mm for any rotated x-y frame
- $D_{xx} = 5.8612 \times 10^4$ N · mm
- $D_{yy} = 1.5187 \times 10^4$ N · mm

Now, confirmed that this laminate satisfies all stiffness thresholds in both the global and rotated coordinate frames. We chose a low volume fraction because achieving the D matrix requirements necessitates greater thickness, which ultimately leads to a lower volume fraction.

The second laminate configuration, noted as **$[0\ 0\ 0\ 45\ 90\ 90\ 45\ 0\ 0\ 0]$** , is specifically optimized for strength. In this arrangement, the volume fraction is set at **0.6**, while each lamina has a thickness of **0.083 mm**. The total weight of the composite is **12.4577 kg /m²**. This design successfully passed all load cases when assessed using the *Tsai-Wu* failure criterion, benefiting from a reduced overall weight due to a combination of fewer plies and a thinner construction. However, it failed to meet the minimum stiffness requirements across all orientations, making it unsuitable for structural applications where rigidity is also essential. The plot for the *Tsai-Wu* criteria for different lamina types designed for strength is attached below as Figure 1.

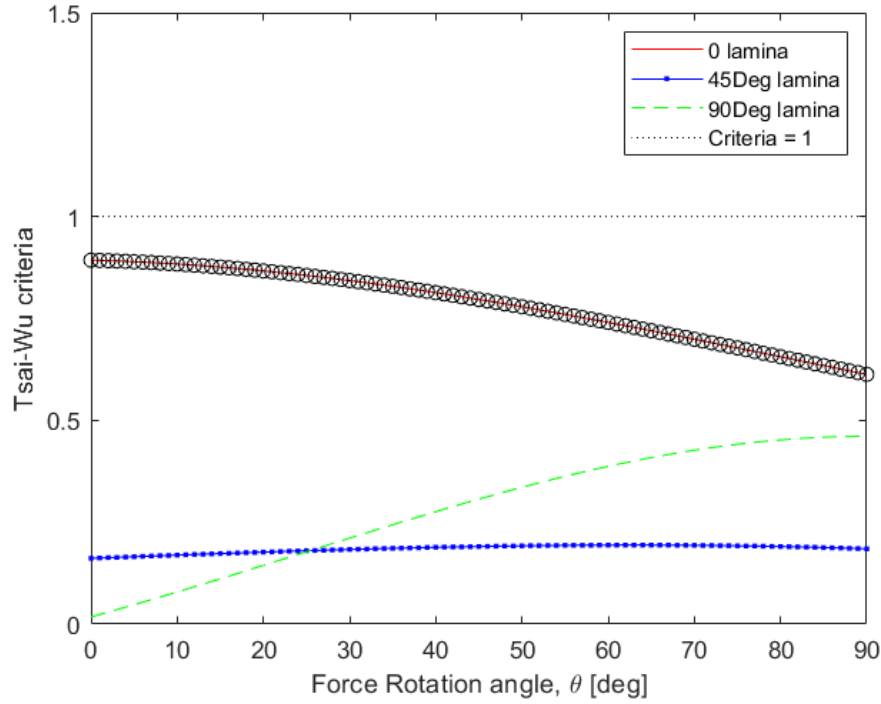


Figure 1: Tsai-Wu criteria

As can be seen from the above figure all the laminas are safe to the given load case and hence our layup has sufficient strength. Now, moving forward the next step is to form a laminate considering both stiffness and strength ability. The final laminate $[0\ 0\ 0\ 0\ 35\ 90\ 90\ 90\ 90]_s$ achieved both stiffness and strength objectives through a symmetric layup with a fiber volume fraction of 40%. This configuration struck an effective balance between mechanical performance and weight efficiency. Since the exact layup already satisfied our stiffness design parameters we only needed to check for strength design constraints now. The thickness **0.125 mm** was adjusted in line with the specified volume fraction-to-thickness relationship. The values for the stiffness design parameters are :

- $A_{xx} = 1.0207 \times 10^5$ N/mm in the global x-y reference frame
- $\min(A_{xx}) = 8.2686 \times 10^4$ N/mm for any rotated x-y frame
- $D_{xx} = 5.8612 \times 10^4$ N · mm
- $D_{yy} = 1.5187 \times 10^4$ N · mm

These values confirm that our laminate satisfies the Stiffness Design considerations. The *Tsai-Wu* criteria plot for types of laminas in the current layup is shown in Figure 2.

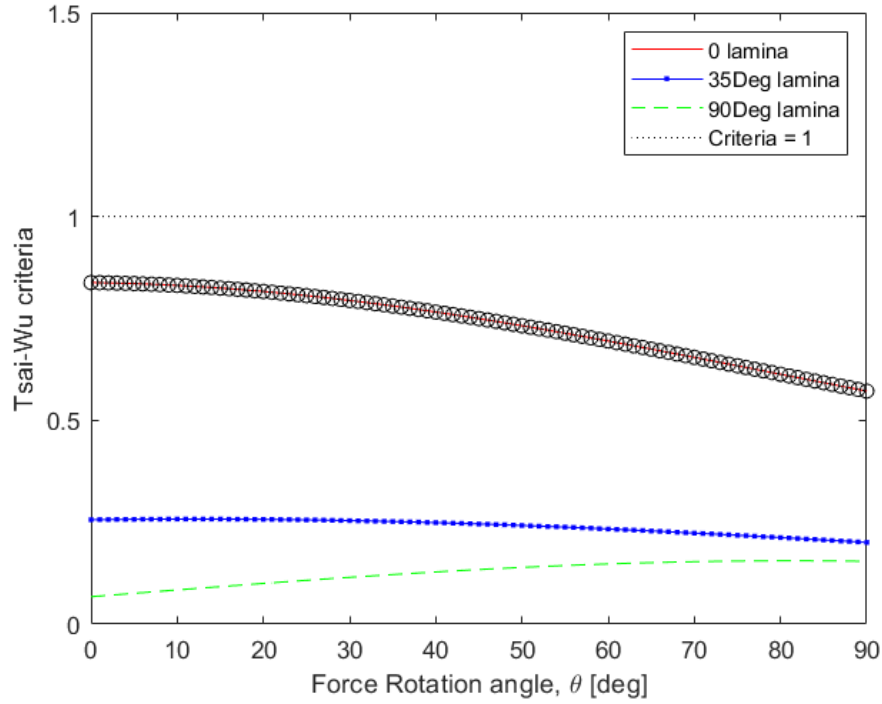


Figure 2: Tsai-Wu criteria for final lamina

As can be seen from the above plot, this configuration successfully met the strength design constraints too. Hence, it satisfies both stiffness and strength evaluations while minimizing structural weight at **27.8604 kg/m²**. As a result, it was ultimately selected as the optimal solution.

4 Conclusion

This study effectively presents an optimized approach to designing composite laminates that successfully balance stiffness and strength while minimizing weight. Through a systematic analysis of the *ABD* matrix and application of the *Tsai-Wu* failure criterion, three distinct laminate configurations were assessed. The final design not only achieved structural integrity but also enhanced weight efficiency, demonstrating that strategic layering and careful material selection are vital. This methodology emphasizes the trade-offs inherent in composite design and provides a solid framework for future advancements in lightweight, high-performance materials.

References

- [1] I.M. Daniel and O. Ishai. *Engineering Mechanics of Composite Materials*. Engineering Mechanics of Composite Materials. Oxford University Press, 1994.
- [2] R.M. Jones. *Mechanics Of Composite Materials*. 500 Tips. Taylor & Francis, 1998.

Appendices

A *Q-S Matrix Code*

```
%% ----- Q-S Matrix ----- %%

function [Q,S,T] = calculate_Q_S_matrix_AKS( fiber_properties,
matrix_properties, composite_properties )

E1f = fiber_properties(1);
E2f = fiber_properties(2);
G12f = fiber_properties(3);
nu12f = fiber_properties(4);

Em = matrix_properties(1);
num = matrix_properties(2);
Gm = Em / (2*(1+num));

Vf = composite_properties(1);
xi1 = composite_properties(2);
xi2 = composite_properties(3);
orientation_angle = composite_properties(4);

m = cos( orientation_angle );
n = sin( orientation_angle );

T = [ m^2 n^2 2*m*n;
      n^2 m^2 -2*m*n;
      -m*n m*n m^2-n^2];

E1 = E1f*Vf + Em*(1 - Vf);
eta1 = (E2f - Em)/(E2f + xi1 * Em );
E2 = Em*(1 +xi1 * eta1 * Vf)/(1 - eta1 * Vf );
nu12 = nu12f * Vf + num * (1 - Vf );
nu21 = nu12 * E2 / E1;
eta2 = (G12f - Gm)/(G12f + xi2 * Gm );
G12 = Gm*(1 + eta2 * xi2 * Vf )/(1 - eta2 * Vf );
```

```

S_123 = [ 1/E1 , -nu21/E2 , 0 ;
        -nu12/E1 , 1/E2 , 0 ;
        0 , 0 , 1/G12/2];

Q11 = E1/(1 - nu12 * nu21 );
Q22 = E2/(1 - nu12 * nu21 );
Q12 = nu21*E1/(1 - nu12 * nu21 );
Q21 = nu12*E2/(1 - nu12 * nu21 );
Q66 = G12;

Q_123 = [ Q11 Q12 0;
        Q21 Q22 0;
        0 0 2*Q66];

S = inv(T)*S_123*T;
Q = inv(T)*Q_123*T;

S(3,:) = S(3,)*2;
Q(:,3) = Q(:,3)/2;

```

B *ABD Matrix Code*

```

%% ----- ABD Matrix ----- %%

function [ABD_matrix] = calculate_ABD_matrix_AKS(Q_combined,t_vector)

A=zeros(3,3);
B=zeros(3,3);
D=zeros(3,3);

total_t = sum(t_vector);
z0 = -total_t/2;
coord_lamina_start = [];

for n=1:length(t_vector)+1

    if n==1
        coord_lamina_start(n) = z0;
    else
        coord_lamina_start(n) = coord_lamina_start(n-1) + t_vector(n-1);
    end
end

```

```

        end
    end

    for i=1:3
        for j=1:3
            for k=1:length(t_vector)

                Q_k = Q_combined( 1:3, ((3*k)-2):3*k );

                A(i,j) = A(i,j) + Q_k(i,j) * (coord_lamina_start(k+1) -
                    coord_lamina_start(k));
                B(i,j) = B(i,j) + (1/2) * Q_k(i,j) * ((coord_lamina_start(k+1))^2 -
                    (coord_lamina_start(k))^2);
                D(i,j) = D(i,j) + (1/3) * Q_k(i,j) * ((coord_lamina_start(k+1))^3 -
                    (coord_lamina_start(k))^3);

            end
        end
    end

    ABD_matrix = [A,B ; B,D];

end

```